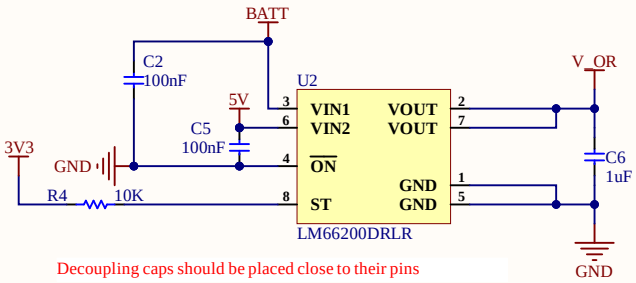


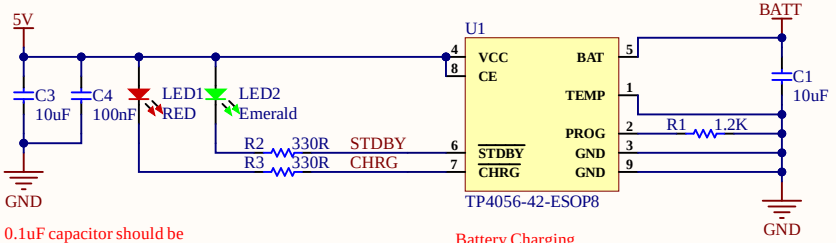
POWER SUPPLY ORING CIRCUIT



Decoupling caps should be placed close to their pins

The highest voltage supply is automatically selected. ST pin is used to detect which voltage is currently selected

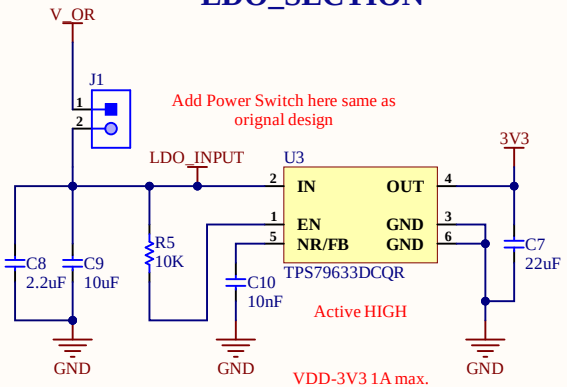
BATTERY CHARGING CIRCUIT

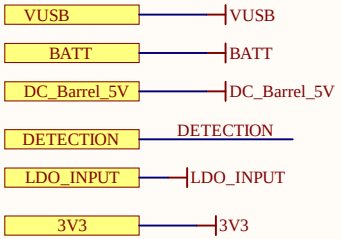


0.1uF capacitor should be placed closest to VCC pin

Battery Charging current = 1000mA

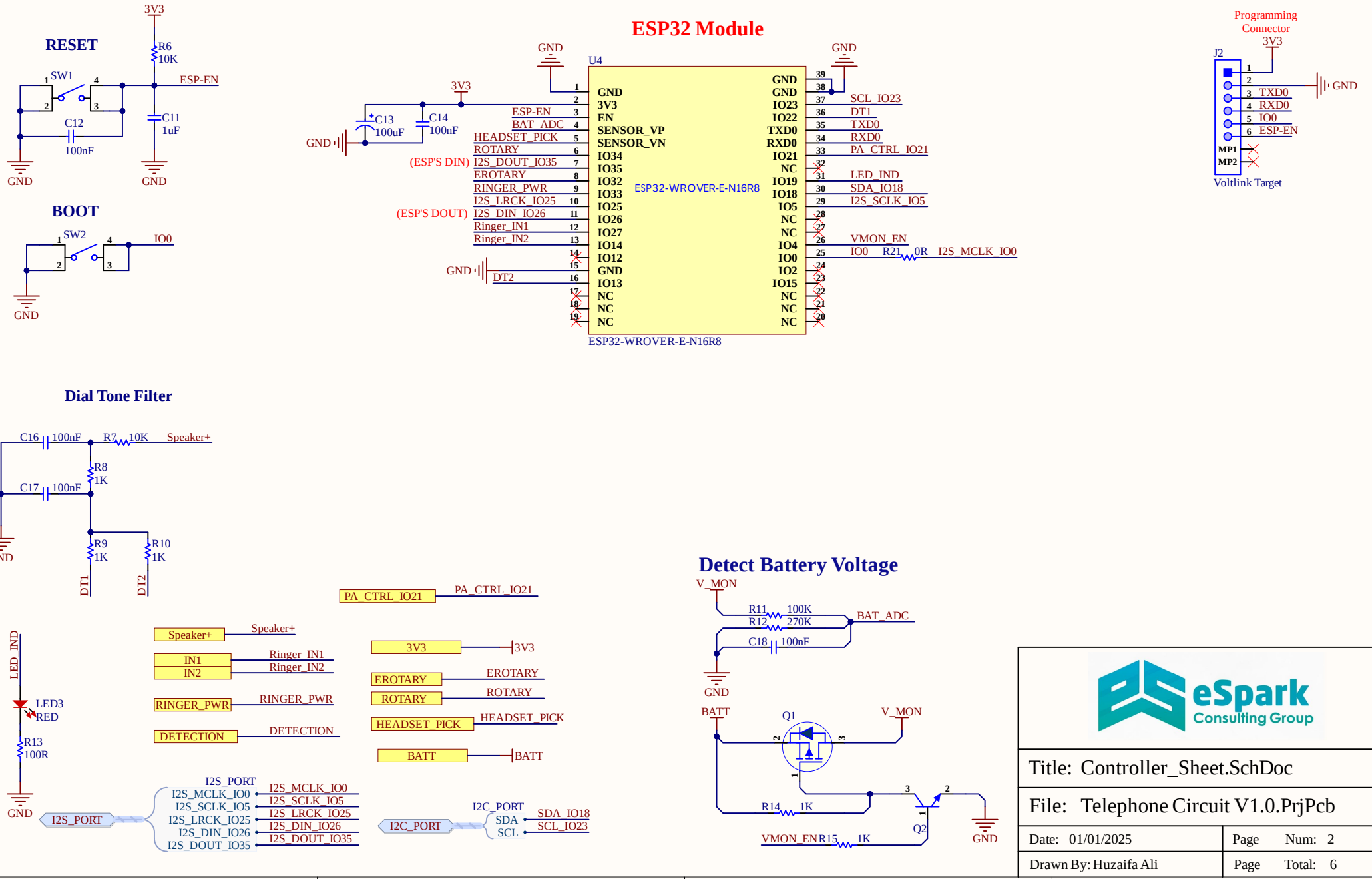
LDO_SECTION

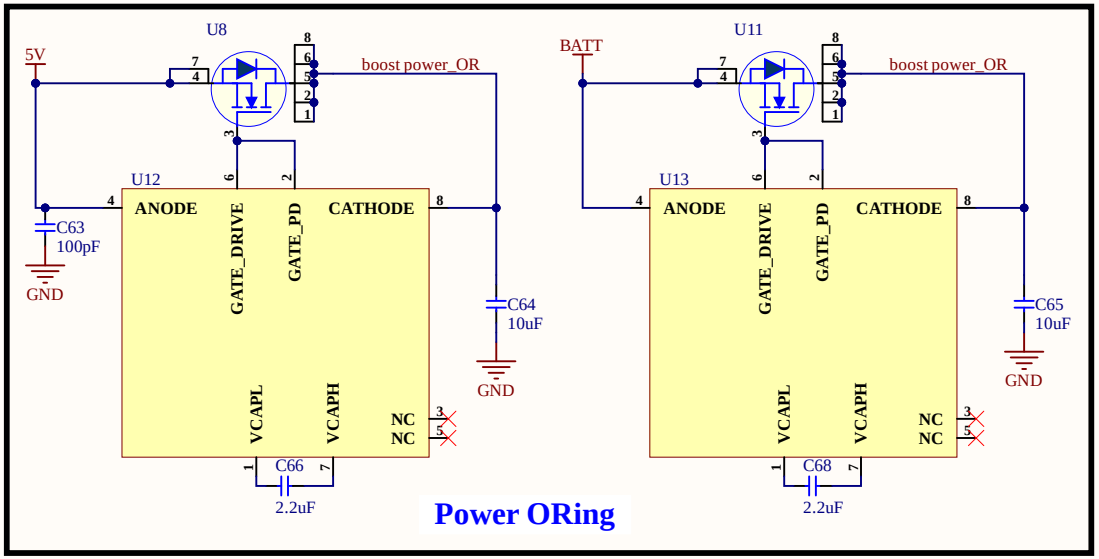


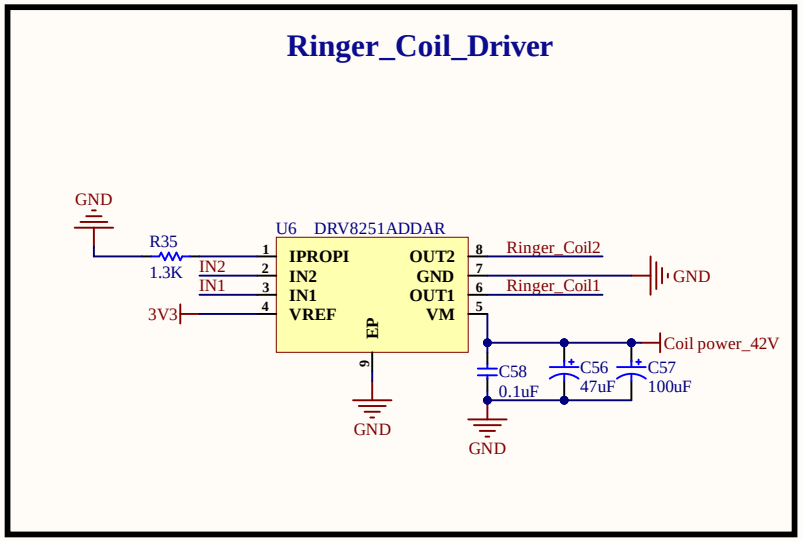


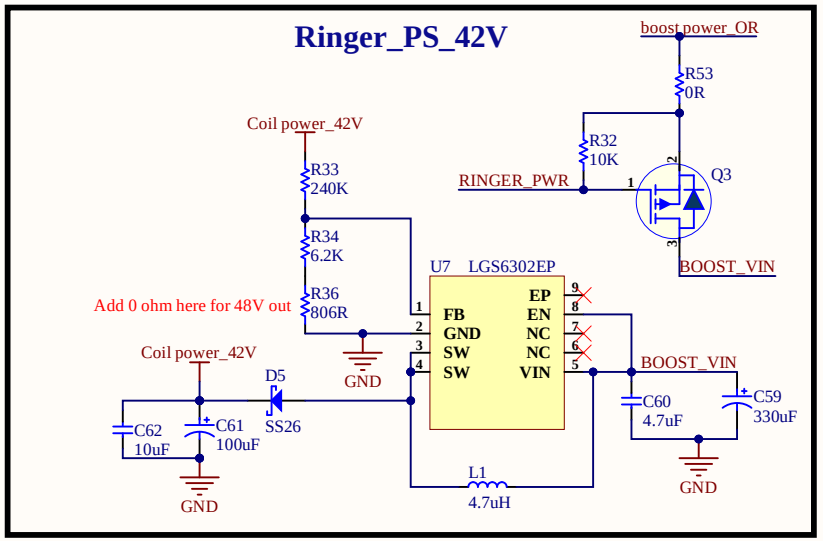
eSpark Consulting Group

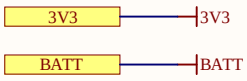
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File: Telephone Circuit V1.0.PrjPcb	
Date: 01/01/2025	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 6

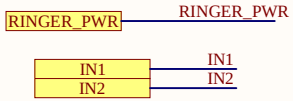

RESET
3V3
R6
10K
SW1
ESP-EN
C12
100nF
C11
1uF
GND
GND
BOOT
SW2
IO0
GND
ESP32 Module
U4
GND
3V3
EN
SENSOR_VP
SENSOR_VN
IO34
IO35
IO32
IO33
IO25
IO26
IO27
IO14
IO12
GND
IO13
NC
NC
NC
GND
GND
IO23
IO22
TXD0
RXD0
IO21
NC
IO19
IO18
IO5
NC
NC
IO4
IO0
IO2
IO15
NC
NC
NC
ESP32-WROVER-E-N16R8
ESP32-WROVER-E-N16R8
C13
100uF
C14
100nF
ESP-EN
BAT_ADC
HEADSET_PICK
ROTARY
(ESP'S DIN) I2S_DOUT_IO35
EROTARY
RINGER_PWR
I2S_LRCK_IO25
(ESP'S DOUT) I2S_DIN_IO26
Ringer_IN1
Ringer_IN2
DT2
SCL_IO23
DT1
TXD0
RXD0
PA_CTRL_IO21
LED_IND
SDA_IO18
I2S_SCLK_IO5
VMON_EN
IO0 R21 0R I2S_MCLK_IO0
Programming Connector
3V3
J2
GND
TXD0
RXD0
IO0
ESP-EN
MP1
MP2
Voltlink Target
Dial Tone Filter
C16 100nF
R7 10K
Speaker+
R8
1K
C17 100nF
R9
1K
R10
1K
DT1
DT2
GND
LED_IND
LED3
RED
R13
100R
GND
Speaker+
IN1
IN2
RINGER_PWR
DETECTION
PA_CTRL_IO21
3V3
EROTARY
ROTARY
HEADSET_PICK
BATT
I2S_PORT
I2S_MCLK_IO0
I2S_SCLK_IO5
I2S_LRCK_IO25
I2S_DIN_IO26
I2S_DOUT_IO35
I2C_PORT
SDA
SCL
SDA_IO18
SCL_IO23
Detect Battery Voltage
V_MON
R11 100K
R12 270K
C18 100nF
BAT_ADC
GND
BATT
Q1
V_MON
R14 1K
VMON_EN
R15 1K
Q2
GND
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Title: Controller_Sheet.SchDoc
File: Telephone Circuit V1.0.PrjPcb
Date: 01/01/2025
Page Num: 2
Drawn By: Huzaifa Ali
Page Total: 6

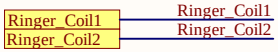








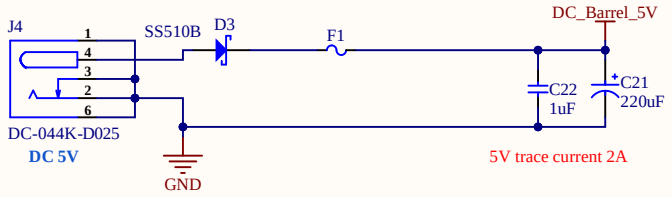




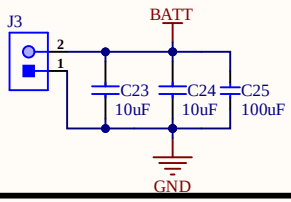
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Date: 01/01/2025	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 6

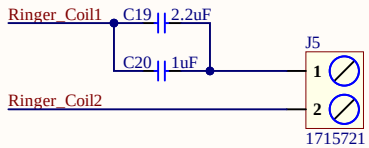
DC BARREL JACK INPUT 5V



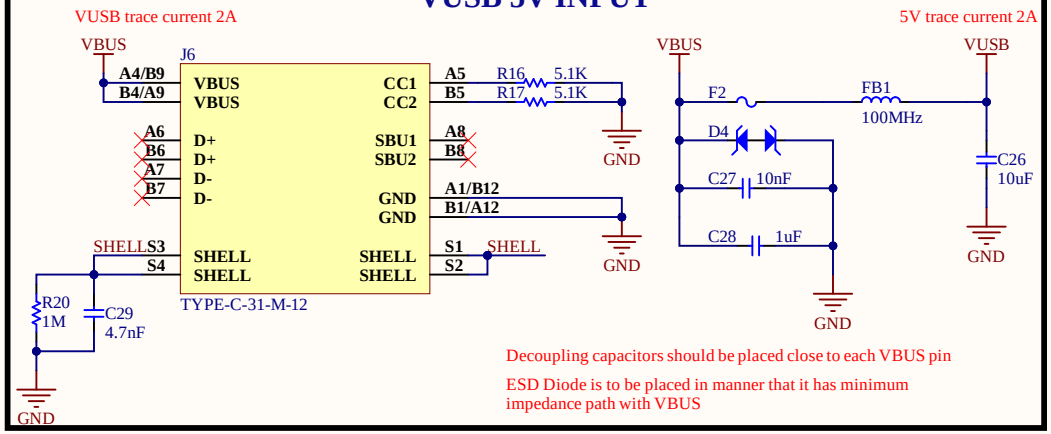
BATTERY INPUT



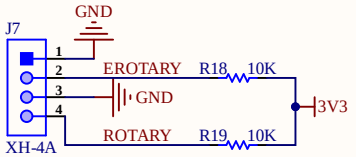
Ringer_Coil



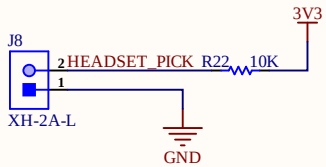
VUSB 5V INPUT



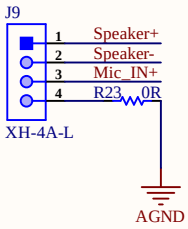
Rotary Dialer

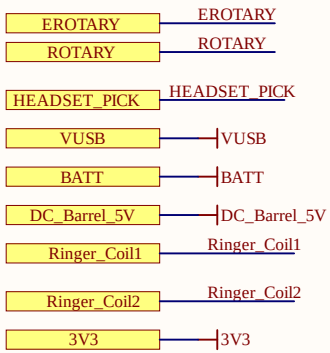


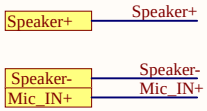
Headset Detection



Headset_Mic and Speaker



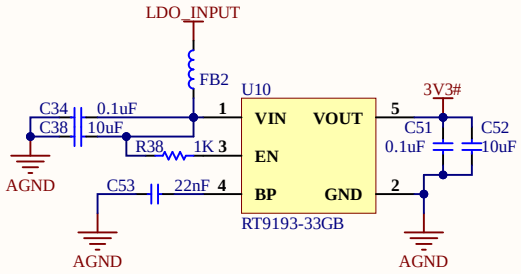




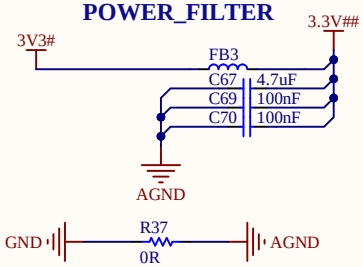
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Date: 30/12/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 6

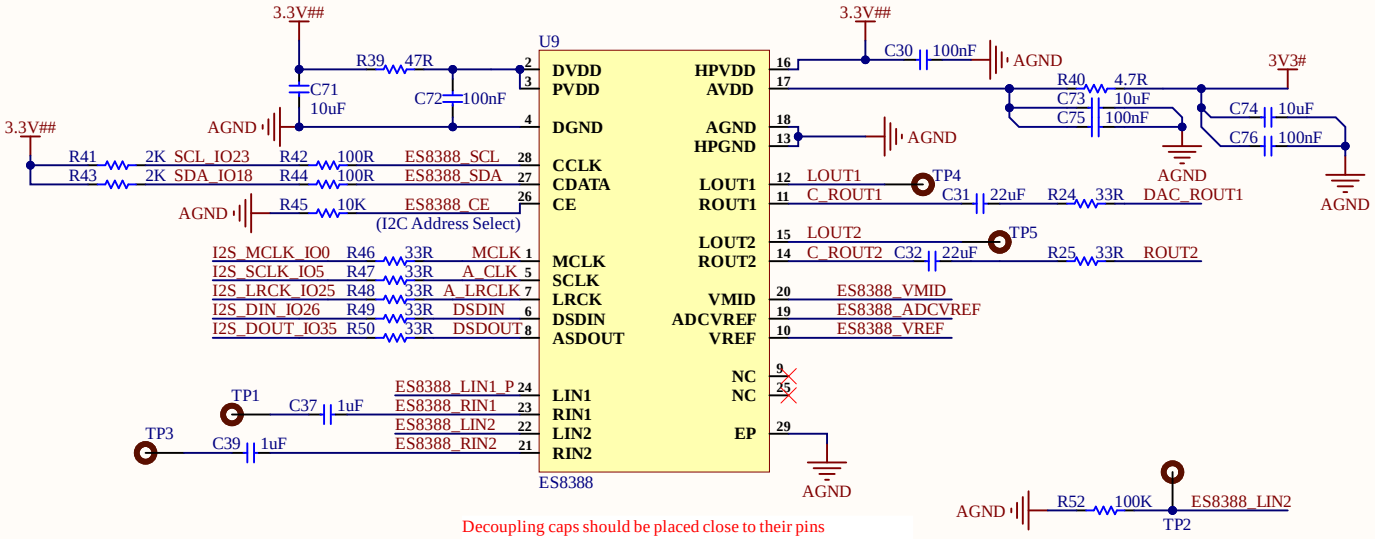
AUDIO_LDO_SECTION



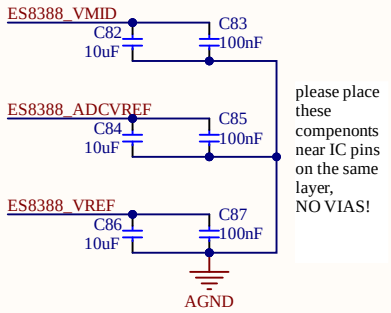
POWER_FILTER



Audio Codec

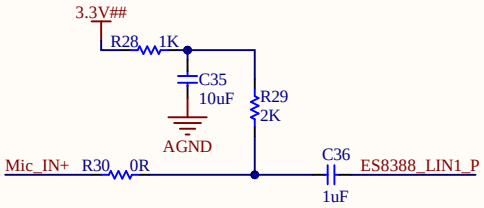


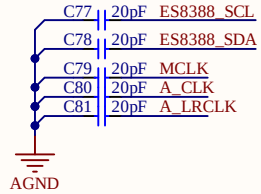
SUPPLY_DECOUPLING

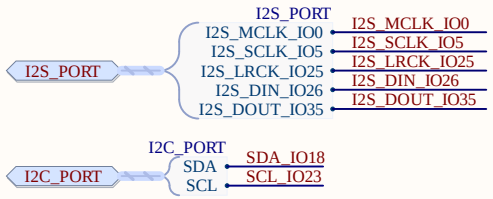


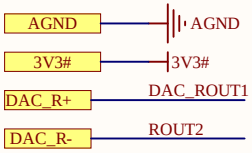
please place these compenonts near IC pins on the same layer, NO VIAS!

MIC Section







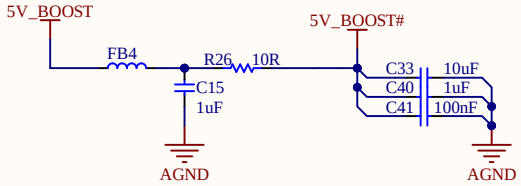


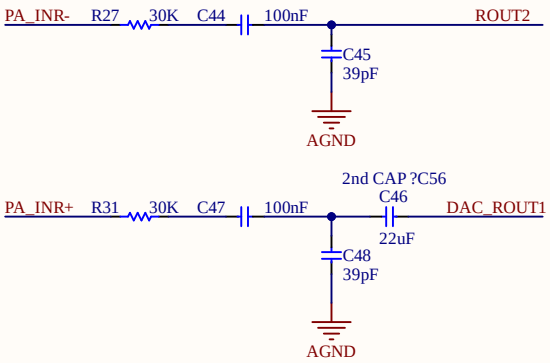


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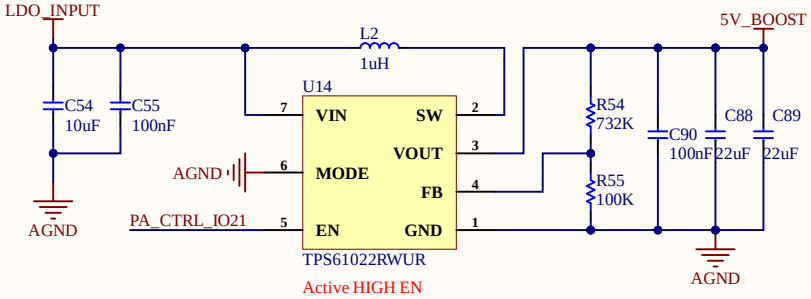
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File: Telephone Circuit V1.0.PrjPcb	
Date: 01/01/2025	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 6

Power_Filter

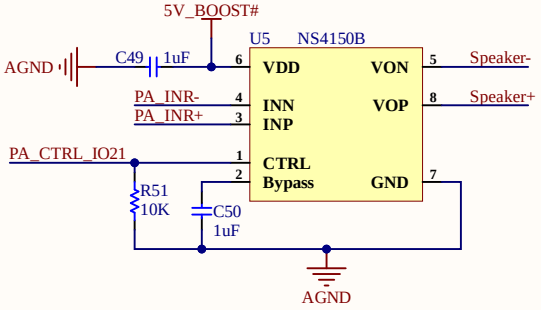


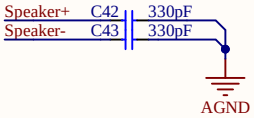


BOOST CONVERTER



Audio Amplifier





LDO_INPUT — LDO_INPUT

AGND — AGND

DAC_R+ — DAC_ROUT1

DAC_R- — ROUT2

Speaker+ — Speaker+

Speaker- — Speaker-

PA_CTRL_IO21 — PA_CTRL_IO21

PA_CTRL_IO21 — PA_CTRL_IO21

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Title: Audio Amplifier_Sheet.SchDoc	
File: Telephone Circuit V1.0.PrjPcb	
Date: 01/01/2025	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 6